Worksheet 3-3 Derivatives of Inverse Functions #1
AP Calculus Name_____

1.) Given $f(x) = 3x^5 + 2x + 3$ and $g(x) = f^{-1}(x)$, find $g'(x)$.

2.) Given $g(x) = f^{-1}(x)$, $f(7) = -2, f(-5) = 1, f'(7) = -1,$ and $f'(-5) = 8$, find $g'(-2)$.

3.) Given $g(x) = f^{-1}(x)$, $f(6) = -8, f(-5) = 1, f'(6) = 3,$ and $f'(-5) = 6$, find $g'(1)$.

The table below gives values of the differentiable functions g and g' at selected values of x.
Let $h(x) = g^{-1}(x)$

x	$g(x)$	$g'(x)$
2	5	3
4	7	−2
5	10	11
7	2	−7
10	4	6

4.) Find equation of the tangent line to $h(x)$ at $x = 2$.

5.) Find equation of the tangent line to $h(x)$ at $x = 7$.

6.) Given $g^{-1}(x) = f(x)$ and $g(x) = \frac{1}{5-x}$, find $f'\left(\frac{1}{3}\right)$.

f and g are differentiable functions. Use the table to answer the problems below. f and g are **NOT** inverses!

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
−1	−2	5	−5	−4
−2	−3	6	−6	−3
−3	−6	−4	−5	−2
−4	−1	−9	−3	−1
−5	−4	−1	−2	−6
−6	−5	−2	−1	−5

7.) Find $g^{-1}(-1)$

8.) Find $f^{-1}(-5)$

9.) Find $\frac{d}{dx}[g^{-1}(x)]_{x=-2}$

10.) Find $\frac{d}{dx}[f^{-1}(x)]_{x=-3}$

TEST PREP

11.) The functions f and g are differentiable for all real numbers and g is strictly increasing. The table below gives values of the functions and their first derivatives at selected values of x. The function h is given by $h(x) = f\big(g(x)\big) - 6$.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	−4	4	2
4	−1	3	6	7

(a) Explain why there must be a value r for $1 < r < 3$ such that $h(r) = -5$.

(b) If g^{-1} is the inverse function of g, write an equation for the line tangent to the graph of $y = g^{-1}(x)$ at $x = 2$.